

Design & Analysis of Algorithms

Lab 1: Algorithm Performance Analysis Workbench

Name: Priyam Sharma

Roll No: 2501730184

Email: 2501730184@krmu.edu.in

Q1. 1.2 Linear Search and Binary Search Comparison (5 marks)

```
def compare_search_algorithms(arr, target):
    linear_index = -1
    linear_comparisons = 0
    for i in range(len(arr)):
        linear_comparisons += 1
        if arr[i] == target:
            linear_index = i
            break

    binary_index = -1
    binary_comparisons = 0
    low = 0
    high = len(arr) - 1

    while low <= high:
        mid = (low + high) // 2
        binary_comparisons += 1

        if arr[mid] == target:
            binary_index = mid
            high = mid - 1
        elif arr[mid] < target:
            low = mid + 1
        else:
            high = mid - 1

    if linear_comparisons < binary_comparisons:
        better_alg = "Linear Search"

    elif binary_comparisons < linear_comparisons:
        better_alg = "Binary Search"

    else:
        better_alg = "Both Equal"

    return [
        "Search Comparison Report ",
        "Linear Search ",
        f"Index: {linear_index}",
        f"Comparisons: {linear_comparisons}",
        "Binary Search",
        f"Index: {binary_index}",
        f"Comparisons: {binary_comparisons}",
        f"Better Algorithm: {better_alg}"
    ]
```

Q2. 1.3 Bubble Sort and Insertion Sort Performance Comparison (10 marks)

```
def compare_bubble_insertion(random_data, sorted_data, reverse_data):
    datasets = [
        ("Random Dataset", random_data),
        ("Sorted Dataset", sorted_data),
        ("Reverse Dataset", reverse_data)
    ]

    report = ["Sorting Performance Report"]

    for dataset_name, arr in datasets:
        b_arr = list(arr)
        n = len(b_arr)
        bubble_comps = 0
        bubble_swaps = 0
```

```

for i in range(n):
    swapped = False
    for j in range(0, n - i - 1):
        bubble_comps += 1
        if b_arr[j] > b_arr[j + 1]:
            b_arr[j], b_arr[j + 1] = b_arr[j + 1], b_arr[j]
            bubble_swaps += 1
            swapped = True
    if not swapped:
        break

i_arr = list(arr)
insertion_comps = 0
insertion_shifts = 0

for i in range(1, len(i_arr)):
    key = i_arr[i]
    j = i - 1
    while j >= 0:
        insertion_comps += 1
        if i_arr[j] > key:
            i_arr[j + 1] = i_arr[j]
            insertion_shifts += 1
            j -= 1
        else:
            break
    i_arr[j + 1] = key

if bubble_comps < insertion_comps:
    better_alg = "Bubble Sort"
elif insertion_comps < bubble_comps:
    better_alg = "Insertion Sort"
else:
    better_alg = "Both Equal"

bubble_str = " ".join(map(str, b_arr))
insertion_str = " ".join(map(str, i_arr))

report.extend([
    dataset_name,
    f"Bubble Sorted: {bubble_str}",
    f"Bubble Comparisons: {bubble_comps}",
    f"Bubble Swaps: {bubble_swaps}",
    f"Insertion Sorted: {insertion_str}",
    f"Insertion Comparisons: {insertion_comps}",
    f"Insertion Shifts: {insertion_shifts}",
    f"Better Algorithm: {better_alg}"
])

return report

```